

6(a)

*three from:*

same / similar, shape as, HMG CoA / substrate ;

complementary (shape) to active site (shape) ;

binds / attaches / fits into / active site (of, enzyme / HMG CoA reductase) ;

HMG CoA / substrate, cannot bind (to active site) ;

**A** no / few, enzyme/substrate complexes form

mevalonic acid production, decreases / stops ;      **I** product not made

AVP ;    e.g. increase in mevinoлин concentration increases inhibition / ora

**3**

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 6(b)     | <p><i>two from:</i></p> <p><i>treatment of disease</i></p> <p><b>I statements about locating or diagnosis of disease</b></p> <p>specific / targeted, therapy / treatment ;</p> <p>    <b>A</b> specific to diseased, cells / tissue</p> <p>    <b>A</b> examples</p> <p>bind to / recognition of, receptors / antigens, on cell surface (of diseased cells) ;</p> <p>kill the cell by stimulating the immune system / AW ;</p> <p>    <b>A</b> correct immune responses e.g. stimulates phagocytosis / activates B-cells</p> <p>attach, radioactive substance / drug (to treat / kill diseased cells) / AW ;</p> <p>treat infectious diseases / bind to (antigens on) pathogens / recognition of antigens on pathogens ;      <b>A</b> viruses</p> <p>    <b>I</b> bacteria <i>unless stated that they cause disease</i></p> <p>    <i>allow examples such as treatment of ebola or rabies or tetanus (by immunising with antibody)</i></p> | 2     |